Examination Question Difficulty and Performance Analysis

Python Programming – Activity-Based Learning (ABL) Assignment

1. Problem Statement

Compare student answer strings against an answer key, calculate question-wise success rates, identify the most frequently missed questions, and classify question performance.

2. Objectives

• Compare each student's answer with the correct answer key.

• Calculate correct and incorrect response counts.

• Find question-wise success percentages.

• Identify frequently missed questions.

• Classify each question by performance.

3. Assumptions

The sample data contains five students and six multiple-choice questions. Answers are compared after trimming whitespace and converting to uppercase. A question is Easy when its success rate is at least 80%, Moderate when it is from 50% to below 80%, and Difficult when it is below 50%. These thresholds are an example policy and can be changed.

4. Algorithm

1. Store the correct answer for each question.

2. Store each student's submitted answers.

3. For every question, compare student answers with the key.

4. Count correct and incorrect responses.

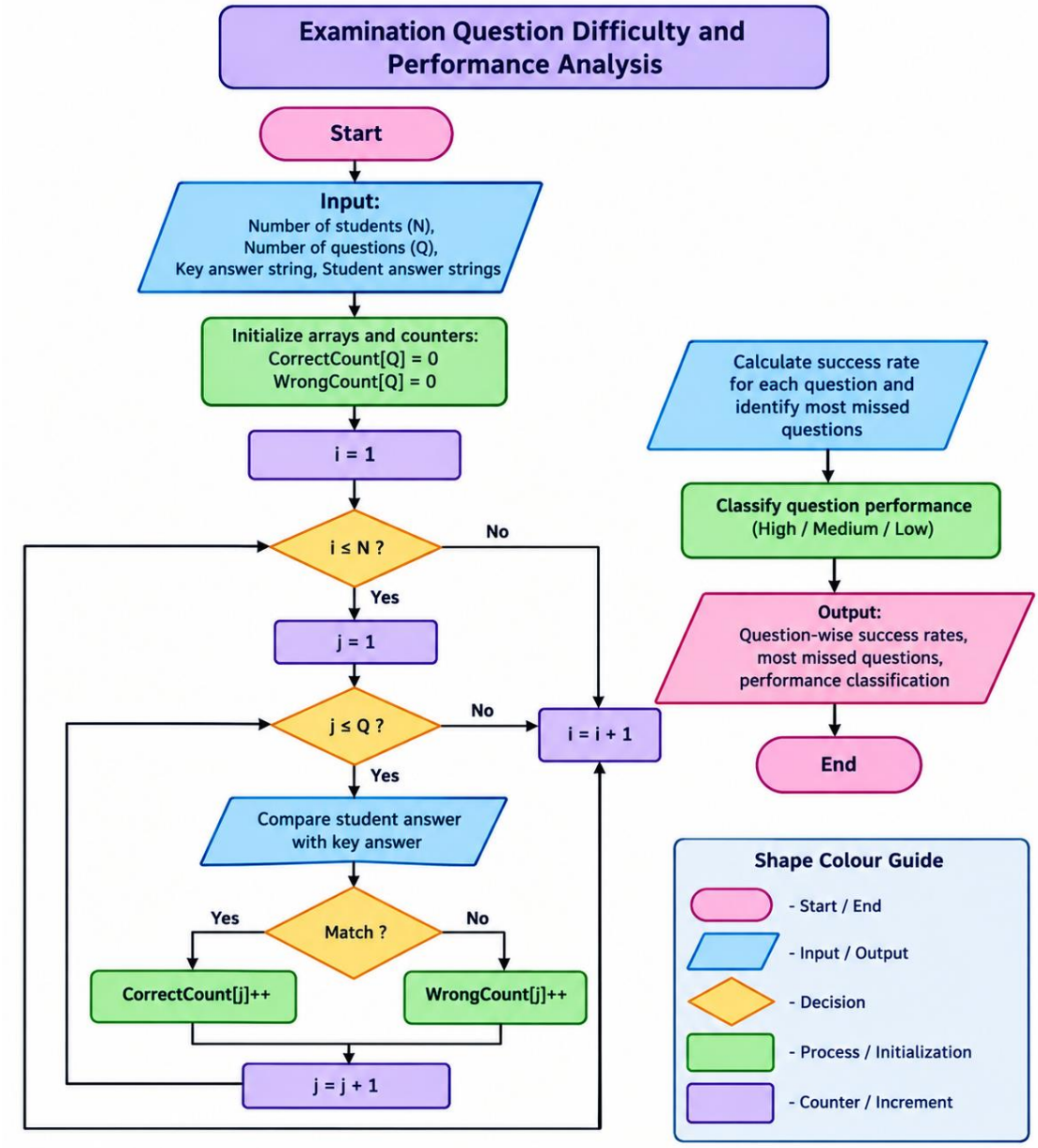
5. Calculate success rate = correct responses / students * 100.

6. Classify the question using the stated thresholds.

7. Sort questions by incorrect response count.

8. Display question statistics and overall performance.

5. Flowchart



6. Pseudocode

BEGIN

INPUT number_of_students (N)

INPUT number_of_questions (Q)

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INPUT key_answer_string
INPUT student_answer_strings

INITIALIZE CorrectCount[Q] = 0
INITIALIZE WrongCount[Q] = 0

FOR i = 1 TO N
  FOR j = 1 TO Q

    IF student_answer[i][j] == key_answer[j] THEN
      CorrectCount[j] = CorrectCount[j] + 1
    ELSE
      WrongCount[j] = WrongCount[j] + 1
    END IF

  END FOR
END FOR

max_wrong = -1

FOR j = 1 TO Q

  success_rate[j] = (CorrectCount[j] / N) * 100

  IF WrongCount[j] > max_wrong THEN
    max_wrong = WrongCount[j]
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    END IF

    IF success_rate[j] >= 75 THEN
      performance[j] = "High"
    ELSE IF success_rate[j] >= 40 THEN
      performance[j] = "Medium"
    ELSE
      performance[j] = "Low"
    END IF

  END FOR

  PRINT "Question-wise Success Rates"
  FOR j = 1 TO Q
    PRINT question_number, success_rate[j], "%"
    PRINT performance[j]
  END FOR

  PRINT "Most Frequently Missed Questions"
  FOR j = 1 TO Q
    IF WrongCount[j] == max_wrong THEN
      PRINT question_number[j], WrongCount[j]
    END IF
  END FOR

END
```

7. Python Source Code

```
# Examination Question Difficulty and Performance Analysis

# Answer key
answer_key = ["A", "B", "C", "D", "A"]

# Student answers
student_answers = [
   ["A", "B", "C", "A", "B"],
   ["A", "C", "C", "D", "A"],
   ["B", "B", "D", "D", "C"],
   ["A", "B", "C", "C", "A"],
   ["C", "A", "B", "D", "B"]
]

total_students = len(student_answers)
total_questions = len(answer_key)

success_rates = []
missed_counts = []

print("=" * 60)
print(" EXAMINATION QUESTION PERFORMANCE ANALYSIS")
print("=" * 60)
```

```
# Analyze each question
for q in range(total_questions):
    correct_count = 0

    for student in student_answers:
        if student[q].upper() == answer_key[q].upper():
            correct_count += 1

    missed_count = total_students - correct_count
    success_rate = (correct_count / total_students) * 100

    success_rates.append(success_rate)
    missed_counts.append(missed_count)

    # Classify question performance
    if success_rate >= 80:
        category = "Easy"
    elif success_rate >= 50:
        category = "Moderate"
    else:
        category = "Difficult"

    print(f"\nQuestion {q + 1}")
    print(f"Correct Answers : {correct_count}")
    print(f"Wrong Answers   : {missed_count}")
```

```
    print(f"Success Rate    : {success_rate:.2f}%")
    print(f"Classification  : {category}")

# Identify the most frequently missed questions
max_missed = max(missed_counts)

print("\n" + "=" * 60)
print(" MOST FREQUENTLY MISSED QUESTIONS")
print("=" * 60)

for q in range(total_questions):
    if missed_counts[q] == max_missed:
        print(f"Question {q + 1}: Missed by {max_missed} students")

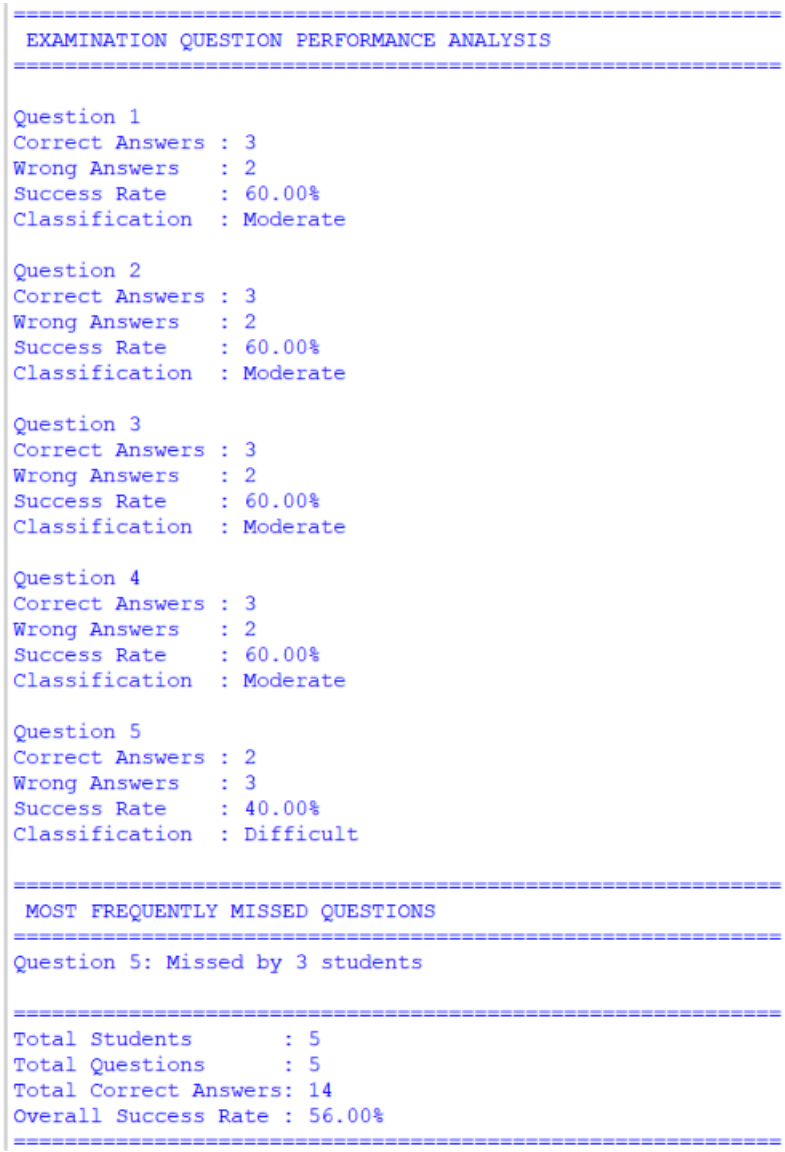
# Overall performance
total_correct = sum(
    1
    for student in student_answers
    for q in range(total_questions)
    if student[q].upper() == answer_key[q].upper()
)

total_answers = total_students * total_questions
overall_rate = (total_correct / total_answers) * 100

print("\n" + "=" * 60)
```

```
print(f"Total Students      : {total_students}")
print(f"Total Questions     : {total_questions}")
print(f"Total Correct Answers: {total_correct}")
print(f"Overall Success Rate : {overall_rate:.2f}%")
print("=" * 60)
```

8. Output Screenshot



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 EXAMINATION QUESTION PERFORMANCE ANALYSIS
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Question 1
Correct Answers : 3
Wrong Answers   : 2
Success Rate    : 60.00%
Classification  : Moderate

Question 2
Correct Answers : 3
Wrong Answers   : 2
Success Rate    : 60.00%
Classification  : Moderate

Question 3
Correct Answers : 3
Wrong Answers   : 2
Success Rate    : 60.00%
Classification  : Moderate

Question 4
Correct Answers : 3
Wrong Answers   : 2
Success Rate    : 60.00%
Classification  : Moderate

Question 5
Correct Answers : 2
Wrong Answers   : 3
Success Rate    : 40.00%
Classification  : Difficult

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 MOST FREQUENTLY MISSED QUESTIONS
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Question 5: Missed by 3 students

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Total Students       : 5
Total Questions      : 5
Total Correct Answers: 14
Overall Success Rate : 56.00%
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```

Test Cases

1. **TC01 – Correct Answers:** Verify that student answers exactly matching the answer key are identified as correct.

2. **TC02 – Incorrect Answers:** Verify that answers differing from the answer key are counted as incorrect.

3. **TC03 – Question-wise Success Rate:** Check whether the success rate for each question is calculated correctly using the number of correct answers and total students.

4. **TC04 – Most Frequently Missed Questions:** Verify that the question or questions with the highest number of incorrect answers are identified.

5. **TC05 – Easy Question Classification:** Verify that questions with a success rate of 80% or above are classified as Easy.

6. **TC06 – Moderate Question Classification:** Verify that questions with a success rate of 50% to below 80% are classified as Moderate.

7. **TC07 – Difficult Question Classification:** Verify that questions with a success rate below 50% are classified as Difficult.

8. **TC08 – Overall Performance:** Check whether the overall success rate is calculated correctly across all students and questions.

9. **TC09 – Multiple Students:** Verify that answers from multiple students are compared against the same answer key correctly.

10. **TC10 – Case-insensitive Comparison:** Check whether lowercase and uppercase answers, such as a and A, are treated as equivalent.

11. **TC11 – All Answers Correct:** Verify that a 100% success rate is displayed when all students answer every question correctly.

12. **TC12 – All Answers Incorrect:** Verify that a 0% success rate is displayed when no student answers any question correctly.

13. **TC13 – Tie in Missed Questions:** Verify that all questions tied for the highest number of incorrect answers are displayed.

14. **TC14 – Single Student:** Check whether the analysis works correctly when only one student's answers are provided.

15. **TC15 – Input Validation:** Verify that invalid inputs, missing answers, or mismatched answer lengths are handled appropriately.

10. Conclusion

The program compares student responses with the answer key, calculates question-wise success rates, highlights the most frequently missed questions, and classifies question performance using a clearly stated policy.